

Day 02 - Git Internals

Objectives:

- Blob, Tree, Commit Objects
- SHA-1
- HEAD & Detached HEAD

Git Objects:

Commit -> Tree -> Blob

Blob:

Stores file contents only.

Tree:

Stores folder structure, filenames, permissions and blob references.

Commit Object:

Tree Reference, Parent Commit, Author, Committer, Timestamp, Commit Message.

HEAD:

Normal: HEAD -> main -> Latest Commit

Detached: HEAD -> Commit

Useful Commands:

```
git log --oneline
git cat-file -p HEAD
git cat-file -p
git checkout
cat .git/HEAD
```

Production:

Detached HEAD helps inspect previous commits without changing branch history.